

Final Project Report - Data and Knowledge Architecture

Fuzzy Logic Implementation: Mamdani and Sugeno

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SC CODE LINK:

<https://github.com/alzidaneMhs/Introduction-To-AI-Final-Project.git>

1. Introduction

This report describes the design and implementation of a Fuzzy Inference System (FIS) used to estimate airline passenger satisfaction based on several inflight service ratings. Two fuzzy methods are implemented from scratch (without using any fuzzy logic library): the Mamdani method and the Sugeno method. Both methods are compared and evaluated using the same dataset and the same rule base.

1.1 Dataset

The dataset used is the Airline Passenger Satisfaction dataset from Kaggle:

<https://www.kaggle.com/datasets/teejmahal20/airline-passenger-satisfaction>

The training data contains 103,904 rows after removing missing values. Each row represents one passenger, with several service ratings (scale 0-5) and a satisfaction label (satisfied / neutral or dissatisfied). For the fuzzy computation, a random sample of 5,000 rows is used, which still satisfies the minimum requirement of 5,000 rows.

1.2 Variables Used

Input variables (5):

1. Inflight wifi service (scale 0-5)
2. Food and drink (scale 0-5)
3. Seat comfort (scale 0-5)
4. Inflight entertainment (scale 0-5)
5. Online boarding (scale 0-5)

Output variable:

- Satisfaction Score (scale 0-100), which is then converted into a label (satisfied / not satisfied) using a threshold of 50, and compared with the ground truth label (satisfaction).

2. Fuzzy Logic Design

2.1 Linguistic Variables

Each input variable is divided into 3 fuzzy sets: Poor, Average, and Good. The output variable (Satisfaction Score) is divided into 3 fuzzy sets: Low, Medium, and High.

Variable	Fuzzy Sets	Range
Inflight Wifi Service	Poor, Average, Good	0 - 5
Food and Drink	Poor, Average, Good	0 - 5
Seat Comfort	Poor, Average, Good	0 - 5
Inflight Entertainment	Poor, Average, Good	0 - 5
Online Boarding	Poor, Average, Good	0 - 5
Satisfaction Score (output)	Low, Medium, High	0 - 100

2.2 Membership Functions

For all input variables (range 0-5), the membership functions are defined as follows:

- Poor: trapezoidal, parameters [0, 0, 1, 2.5]
- Average: triangular, parameters [1, 2.5, 4]
- Good: trapezoidal, parameters [2.5, 4, 5, 5]

For the output variable Satisfaction Score (range 0-100):

- Low: trapezoidal, parameters [0, 0, 30, 50]
- Medium: triangular, parameters [30, 50, 70]
- High: trapezoidal, parameters [50, 70, 100, 100]

The triangular function (trimf) and trapezoidal function (trapmf) are implemented manually as plain Python functions, following the standard formulas for triangular and trapezoidal membership functions.

2.3 Rule Base

There are 20 rules in total (the minimum requirement is 15 rules). Each rule combines the 5 input variables using the AND (minimum) operator, and the consequent is one of Low, Medium, or High. A summary of the rule base is shown below.

No	wifi	food	seat	entertainment	boarding	Consequent
1	good	good	good	good	good	High
2	good	good	good	average	good	High
3	good	average	good	good	average	High
4	average	good	good	good	good	High
5	good	good	average	good	average	High
6	average	average	average	average	average	Medium
7	average	average	good	average	average	Medium

8	good	poor	average	average	average	Medium
9	average	good	poor	average	good	Medium
10	average	average	average	good	poor	Medium
11	poor	average	average	average	average	Medium
12	average	poor	average	poor	average	Low
13	poor	poor	poor	poor	poor	Low
14	poor	poor	average	poor	average	Low
15	poor	average	poor	average	poor	Low
16	average	poor	poor	poor	average	Low
17	poor	poor	poor	average	poor	Low
18	good	poor	poor	poor	poor	Medium
19	average	average	poor	poor	poor	Low
20	good	good	poor	average	poor	Medium

3. Mamdani Fuzzy Inference

The Mamdani method consists of four steps:

6. Fuzzification: each input value (0-5) is converted into membership degrees for Poor, Average, and Good using the membership functions above.
7. Inference: for each rule, the firing strength is calculated as the minimum of the membership degrees of all antecedents (AND = min). Then, implication is applied by clipping the output set of the consequent at the firing strength level (min).
8. Aggregation: the implication results of all fired rules are combined using the maximum operator (OR = max), producing one combined output curve.
9. Defuzzification: the crisp output value is obtained using the centroid (center of gravity) method, calculated over a discretized output range of 0 to 100.

If no rule is fired for a given input (firing strength = 0 for all rules), the system returns a default neutral score of 50.

4. Sugeno Fuzzy Inference

The Sugeno method (order-1) uses the same fuzzification step and the same rule base as Mamdani, but the consequent of each rule is a linear function of the inputs instead of a fuzzy set.

The linear function for each consequent label is defined as follows (base = sum of the 5 input ratings multiplied by 4, giving a base value in the range 0-100):

Consequent	Linear Function z
Low	$z = 4 \times (\text{wifi} + \text{food} + \text{seat} + \text{entertainment} + \text{boarding})$
Medium	$z = 4 \times (\text{wifi} + \text{food} + \text{seat} + \text{entertainment} + \text{boarding}) + 15$
High	$z = 4 \times (\text{wifi} + \text{food} + \text{seat} + \text{entertainment} + \text{boarding}) + 30$, capped at 100

The final output is calculated using the weighted average formula:

$$z^* = (\text{sum of } \alpha_i * z_i) / (\text{sum of } \alpha_i)$$

where α_i is the firing strength of rule i (calculated the same way as in Mamdani, using the AND = min operator). If the sum of all firing strengths is 0, the system returns a default neutral score of 50.

5. Comparison and Evaluation

5.1 Difference in Output (Mamdani vs Sugeno)

Both methods were run on the same 5,000-row sample. The continuous satisfaction scores produced by Mamdani and Sugeno were compared directly using MAE, MSE, and correlation. The results are as follows:

Metric	Value
MAE (Mamdani vs Sugeno)	5.783
MSE (Mamdani vs Sugeno)	108.332
Correlation	0.911
Mean difference (Mamdani - Sugeno)	-5.75

The correlation of 0.911 shows that the two methods generally agree with each other: when the input ratings are higher, both methods tend to produce a higher satisfaction score, and vice versa. However, there is a systematic difference - on average, the Mamdani score is about 5.75 points lower than the Sugeno score for the same input, mainly because of the difference between centroid defuzzification and weighted average defuzzification.

5.2 Evaluation Against Ground Truth

Since the dataset has a ground truth label (satisfaction), the fuzzy output score is converted into a label (satisfied if score ≥ 50 , otherwise not satisfied) and compared with the ground truth using accuracy, precision, recall, and F1-score.

Method	Accuracy	Precision	Recall	F1-Score
Mamdani	0.587	0.514	0.987	0.676
Sugeno	0.560	0.498	0.989	0.662

Both methods achieve a high recall (around 0.99), meaning they rarely miss passengers who are actually satisfied. However, the precision is lower (around 0.50), meaning a fairly large number of passengers who are predicted as 'satisfied' are actually not satisfied according to the ground truth. The overall accuracy is around 0.56 - 0.59 for both methods, with Mamdani slightly higher than Sugeno.

6. Interpretation

6.1 Differences Between Mamdani and Sugeno

- Mamdani produces its output through aggregation of fuzzy output sets and centroid defuzzification. This makes the output curve smoother, but the resulting score tends to be pulled toward the middle of the output range and rarely reaches extreme values (close to 0 or 100).

- Sugeno produces its output directly from a linear function of the inputs combined with a weighted average. This makes the output more proportional to the input values and computationally simpler, since it does not require discretizing or integrating over the output range.
- The high correlation (0.911) between the two methods shows that they generally agree on the overall trend, but the consistent gap (Mamdani lower than Sugeno on average) reflects the structural difference between the two defuzzification methods.

6.2 Advantages and Disadvantages

Mamdani:

- Advantage: more intuitive, since the output is represented as a fuzzy set that can be interpreted linguistically (e.g. Low, Medium, High).
- Advantage: well suited for systems based on expert knowledge, where the consequent naturally takes the form of a linguistic label.
- Disadvantage: defuzzification (centroid) is computationally more expensive, since it requires discretizing and integrating over the output domain.
- Disadvantage: the output tends to be 'pulled' toward the middle of the output range (smoothing effect).

Sugeno:

- Advantage: computationally efficient, since the output is calculated directly from a formula and a weighted average, without needing to process an output fuzzy set.
- Advantage: the output is more precise and proportional to the combination of input values.
- Disadvantage: the consequent is a mathematical function rather than a linguistic label, so the output is harder to interpret in plain language.
- Disadvantage: the parameters of the linear consequent function need to be carefully designed; if they are not representative, the output can become biased.

6.3 Conclusion

Both the Mamdani and Sugeno methods were successfully implemented from scratch and produce satisfaction score estimates that are consistent with each other (correlation 0.911), using the same rule base of 20 rules. When evaluated against the ground truth label, both methods achieve similar overall accuracy (around 0.56 - 0.59), with high recall but moderate precision. Mamdani is more suitable when interpretability and a linguistic representation of the output are important, while Sugeno is more suitable when computational efficiency and proportional, numerical output are the priority.